A Per-Pixel Thermal Model for Gale Crater, Mars, Calibrated to REMS: Surface and Air Temperature Maps and a Dust Devil Heating Index

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Key Points

- A vectorised, implicit 1D thermal model with three fitted global parameters is run for 529 808 pixels of Gale Crater and corrected by a clamped ridge regression fitted to REMS.
- Under leave-one-block-out validation the ground model improves on a seasonal REMS lookup (4.86 vs 5.38 K) but not on a sealed test (5.25 vs 5.06 K); skill away from the traverse is unmeasured.
- REMS behaves like ground of 330–400 tiu everywhere along the traverse, whatever THEMIS reports; a Dust Devil Heating Index built on the same geometry tracks the in-situ ground–air gap but is not yet tested against dust devils.

Abstract

We describe a one-dimensional thermal model built to map surface and near-surface air temperature across Gale Crater, Mars, and an index of solar-driven surface heating derived from the same geometry. The model solves the heat-conduction equation in a homogeneous regolith column below a surface energy balance with parameterised direct, diffuse and atmospheric infrared forcing. Layers thicken geometrically from $\delta/30$ to 11δ, where δ is the diurnal skin depth; time stepping is fully implicit, with a Newton solve of the surface balance; and a deep-temperature shift reduces spin-up from about 60 to 11–14 sols. The solver reproduces the analytic half-space solution to within 0.8% in amplitude and conserves energy to $2 \times 10^{-4}\,\mathrm{W\,m^{-2}}$. Three global parameters (diffuse fraction, sky-infrared fraction and a scale on THEMIS thermal inertia) are fitted to the Curiosity rover's REMS ground temperature, and a ridge regression on a functional-link basis corrects the remaining error. All models are scored by leave-one-block-out (LOBO) validation over nine blocks of contiguous sols, against a benchmark that looks up the mean REMS value by time of day and season. The physics alone scores 6.72 K macro RMSE (benchmark 5.38 K); with the correction, 4.86 K. On two sealed blocks scored once, the model scores 5.25 K against the benchmark's 5.06 K. Air temperature is level with its benchmark (4.10 vs 4.19 K sealed). The maps cover a 718×738 grid at about 200 m for five seasons at 30-minute steps. The Dust Devil Heating Index scores absorbed sunlight divided by $\sqrt{I}$ per 100 m pixel and per sol, normalised to one fixed reference sol. It correlates at 0.80–0.98 with the REMS ground–air gap within sols, and dust suppresses it as expected. All REMS data come from an area of about 10×5 km, so every score here measures skill in time near the rover, not across the crater.

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1 Introduction

Numerical thermal models link the thermal infrared signal of a planetary surface to its physical properties. On Mars the KRC model [14] has been the reference implementation: it solves subsurface conduction below a surface energy balance with an explicit one-layer atmosphere, and has been used in the analysis of observations from almost every Mars mission with infrared sensors and in the selection of almost every landing site [14]. Thermal inertia maps from orbit, such as the THEMIS 100 m mosaic [2, 4], are themselves products of such models.

This paper addresses a narrower forward problem. Given those orbital maps and the in-situ record of the Curiosity rover's REMS ground temperature sensor [6, 9], we want ground and air temperature at every 200 m pixel of Gale Crater, at any local time, for a few representative seasons, together with an honest statement of how much of that map the data can check. The intended use is a surface-operations "survival guide": where and when the ground is coldest, where it heats fastest, and where convective dust lifting is favoured.

Three requirements set the design apart from KRC:

1. **Throughput.** About 530 000 columns per season must run in hours on a laptop, so the solver is vectorised over columns and depends only on NumPy.
2. **Calibration to one in-situ record.** Where KRC computes the atmosphere from physical principles, we parameterise it with two coefficients and fit them, plus a thermal-inertia scale, to REMS. A statistical residual then corrects what the physics misses, in the spirit of model-discrepancy calibration [13]:

$$T(\mathbf{x}, t) = T_{\text{base}}(\mathbf{x}, t; \boldsymbol{\theta}) + f\big(\mathbf{c}(\mathbf{x}), \text{LTST}, L_{\text{s}}, \tau\big). \tag{1}$$

3. **Validation that cannot leak.** Neighbouring sols share weather, dust and rover position, so all scores hold out whole blocks of sols [25].

Section 2 describes the model in the manner of [14]: notation, insolation, atmosphere, boundary conditions, layering and numerics. Section 3 verifies the solver. Section 4 calibrates and validates against REMS. Sections 5 and 6 describe the two applications: the temperature maps and the Dust Devil Heating Index. Section 7 compares the model with KRC and states its limits.

2 Model description

2.1 Notation

Table 1 lists the symbols, units and adopted values. Thermal inertia is in tiu $\equiv \text{J m}^{-2}\,\text{K}^{-1}\,\text{s}^{-1/2}$.

2.2 Orbital geometry and insolation

Solar declination $\delta_\odot$, heliocentric distance r and top-of-atmosphere flux S_0 follow from L_{s}:

$$\sin\delta_\odot = \sin\epsilon_o \sin L_{\text{s}}, \qquad r = \frac{a(1-e^2)}{1 + e\cos(L_{\text{s}} - L_{\text{s}}^{p})}, \qquad S_0 = \frac{1361}{r^2}\ \text{W m}^{-2}, \tag{2}$$

with r in AU. With hour angle $h = 15^\circ(\text{LTST} - 12)$,

$$\mu_0 = \sin\phi\sin\delta_\odot + \cos\phi\cos\delta_\odot\cos h, \tag{3}$$

$$\cos i = \max\big(0,\ \cos s\,\mu_0 + \sin s\,\sin\zeta\,\cos(\alpha_\odot - \alpha_s)\big), \tag{4}$$

where $\alpha_\odot$ is the solar azimuth. All shortwave terms vanish for $\mu_0 \le 0$. Shadowing by distant terrain is not modelled. Each sol is run at fixed L_{s} and τ.

Table 1: Symbols, units and adopted values. Fitted parameters are refitted inside every validation fold.

Symbol	Meaning	Units	Value
P	length of one sol	s	88 775.244
Δt	time step	s	$P/960 \approx 92.5$
ρc	volumetric heat capacity	$\mathrm{J\,m^{-3}\,K^{-1}}$	1.2×10^6
I	thermal inertia, $I = s_{\mathrm{TI}} I_{\mathrm{THEMIS}}$	tiu	per pixel
k	thermal conductivity, $k = I^2/\rho c$	$\mathrm{W\,m^{-1}\,K^{-1}}$	per pixel
κ	thermal diffusivity, $k/\rho c$	$\mathrm{m^2\,s^{-1}}$	per pixel
δ	diurnal skin depth, $\sqrt{kP/\pi\rho c}$	m	0.02–0.10
A	bolometric albedo (TES)	–	per pixel
ε	emissivity	–	0.95 (fixed)
σ	Stefan–Boltzmann constant	$\mathrm{W\,m^{-2}\,K^{-4}}$	$5.670\,374 \times 10^{-8}$
ϵ_o	obliquity	deg	25.19
e, a	eccentricity, semi-major axis	–, AU	0.0934, 1.5237
L_{s}^{p}	L_{s} of perihelion	deg	251
ϕ, s, α_s	latitude, slope, aspect (downslope azimuth, clockwise from north)	deg	per pixel
μ_0	cosine of solar zenith angle ζ	–	computed
τ	column dust optical depth (Mastcam 880 nm)	–	per sol
g_{diff}	fraction of scattered sunlight reaching the surface	–	fitted, $[0, 1]$
f_{IR}	sky infrared as fraction of noon flat sunlight	–	fitted, $[0, 0.15]$
b_τ	growth of sky infrared with τ	–	1 (fixed)
s_{TI}	scale on THEMIS thermal inertia	–	fitted, $[0.7, 1.3]$

2.3 Atmosphere

KRC represents the atmosphere as a single energy-conserving layer with delta-Eddington scattering [14]. We replace it with three closed-form terms whose coefficients are fitted. The direct beam is attenuated along the slant path,

$$Q_{\mathrm{dir}} = S_0 \cos i \; e^{-\tau/\mu_0}. \tag{5}$$

A fraction g_{diff} of the light removed from the beam reaches the surface as isotropic diffuse light, weighted by the sky view factor:

$$Q_{\mathrm{dif}} = g_{\mathrm{diff}}\, S_0\, \mu_0 \left(1 - e^{-\tau/\mu_0}\right) \tfrac{1}{2}(1 + \cos s). \tag{6}$$

Downwelling thermal infrared is held constant within a sol and scales with the noon insolation and the dust load:

$$L_{\mathrm{sky}} = f_{\mathrm{IR}}\, S_0\, \mu_{0,\mathrm{noon}} \left(1 + b_\tau \tau\right) \tfrac{1}{2}(1 + \cos s), \tag{7}$$

where $\mu_{0,\mathrm{noon}}$ is μ_0 at noon on a flat surface. The atmosphere has no heat capacity of its own, and sensible heat exchange with the surface is omitted.

2.4 Surface boundary condition

The surface temperature T_s satisfies

$$(1 - A)\,(Q_{\mathrm{dir}} + Q_{\mathrm{dif}}) + \varepsilon L_{\mathrm{sky}} - \varepsilon\sigma T_s^4 + k\,\frac{\partial T}{\partial z}\Big|_{z=0} = 0. \tag{8}$$

Emissivity is fixed: at night ε and L_{sky} trade off almost exactly, so they are never fitted together. CO_2 frost is not modelled; the coldest temperature anywhere on the map is 151.5 K, above the CO_2 frost point at Gale's surface pressure.

2.5 Subsurface: properties and layering

Within the column,

$$\rho c\,\frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(k\,\frac{\partial T}{\partial z}\right), \qquad \left.\frac{\partial T}{\partial z}\right|_{z_{\max}} = 0, \tag{9}$$

with ρc and k constant in depth and temperature. KRC allows two material zones and temperature-dependent properties; our column is homogeneous.

Like KRC, we use layers that thicken geometrically with depth [14]. Thicknesses are fixed in units of δ: $\Delta z_j = (\delta/30)\,\gamma^{j-1}$ with $\gamma = 1.10$, for $j = 1, \ldots, N$, with N the smallest integer giving a total depth of at least 10δ:

$$N = \left\lceil \frac{\ln\!\left(1 + 30\cdot 10\,(\gamma - 1)\right)}{\ln \gamma} \right\rceil = 37, \qquad z_{\max} = \frac{\delta}{30}\,\frac{\gamma^N - 1}{\gamma - 1} = 11.0\,\delta. \tag{10}$$

Because the grid is identical in skin-depth units for every pixel, any batch of columns shares the same number of nodes. Table 2 gives selected depths.

Table 2: Selected layers in units of δ (left), and the skin depth for representative thermal inertias (right).

Layer j	$\Delta z_j/\delta$	bottom of layer, z_j/δ	I (tiu)	δ (cm)
1	0.033	0.033	150	2.1
5	0.049	0.204	208	2.9
10	0.079	0.531	281	3.9
20	0.204	1.91	361	5.1
30	0.529	5.48	470	6.6
37	1.030	11.00	682	9.6

The specification originally used $\gamma = 1.15$ (28 layers). It failed the analytic amplitude test at 2δ by -1.18% (limit 1%). Halving every layer reduced the error to -0.54%, whereas a four-times-smaller time step only reached -0.94%, so the error was spatial and γ was reduced to 1.10.

2.6 Discretisation

Nodes sit at layer boundaries $z_0 = 0, z_1, \ldots, z_N$. Each node owns half of each adjacent layer, so its heat capacity per unit area is

$$C_0 = \tfrac{1}{2}\rho c\,\Delta z_1, \qquad C_i = \tfrac{1}{2}\rho c\,(\Delta z_i + \Delta z_{i+1})\ (0 < i < N), \qquad C_N = \tfrac{1}{2}\rho c\,\Delta z_N, \tag{11}$$

and layer j has conductance $g_j = k/\Delta z_j$. Placing the surface node at $z = 0$ with half-cell storage gives second-order accuracy and exact discrete energy conservation. Backward Euler gives, for nodes $i \geq 1$,

$$\frac{C_i}{\Delta t}(T_i^{n+1} - T_i^n) = g_i(T_{i-1}^{n+1} - T_i^{n+1}) + g_{i+1}(T_{i+1}^{n+1} - T_i^{n+1}), \tag{12}$$

with $g_{N+1} = 0$ at the bottom. The tridiagonal system is eliminated from the bottom up, giving the recurrence $T_i^{n+1} = \alpha_i + \beta_i T_{i-1}^{n+1}$, in which β_i depends only on the grid and is computed once per run. The surface node then satisfies one scalar equation,

$$R(T_0) = F_{\text{in}} - \varepsilon\sigma T_0^4 + g_1(\alpha_1 + (\beta_1 - 1)\,T_0) - \frac{C_0}{\Delta t}(T_0 - T_0^n) = 0, \tag{13}$$

where $F_{\text{in}} = (1 - A)Q_{\text{sw}} + \varepsilon L_{\text{sky}}$ at the new time level. It is solved by Newton's method,

$$T_0 \leftarrow T_0 - \frac{R(T_0)}{R'(T_0)}, \qquad R'(T_0) = -4\varepsilon\sigma T_0^3 + g_1(\beta_1 - 1) - \frac{C_0}{\Delta t}, \tag{14}$$

to $|R| < 1\times10^{-6}\,\mathrm{W\,m^{-2}}$; failure to converge within 20 iterations raises an error. Back-substitution recovers the profile. Every operation is vectorised over a batch of columns, arrays of shape $(n_{\mathrm{col}}, N+1)$, in float64.

2.7 Time step

KRC integrates explicitly and doubles the time step repeatedly with depth as layers thicken [14]. The implicit scheme here is unconditionally stable, so one step serves all layers: $\Delta t = P/960$, ten steps per 15-minute REMS bin. Accuracy rather than stability sets Δt, and Section 3 checks it directly.

2.8 Spin-up

Each column starts at the radiative-equilibrium temperature of the sol-mean absorbed flux, $\bar{T} = (\overline{F_{\mathrm{in}}}/\varepsilon\sigma)^{1/4}$, and repeats the sol until the largest change between consecutive sols is below 0.01 K at every depth. Unaccelerated, this took about 60 sols: the slowest mode of an 11δ column decays over roughly 13 sols, and the deep layers start 1–4 K from the periodic mean. After each unconverged sol we therefore add a constant to the whole column so that its deepest node equals the sol-mean surface temperature. With zero bottom flux, the time-mean temperature in the periodic state is the same at every depth, so the shift is exact at convergence and vanishes there. It reduces spin-up to 11–14 sols at every tested pixel and lands closer to the converged state (0.013 K) than unaccelerated iteration (0.03 K).

2.9 Fitting the global parameters

The parameters $\boldsymbol{\theta} = (g_{\mathrm{diff}}, f_{\mathrm{IR}}, s_{\mathrm{TI}})$ minimise the sum of squared errors against REMS ground temperature over a fold's fitting rows. The search evaluates a $5\times5\times5$ grid spanning the bounds, then takes up to four projected Gauss–Newton steps with a forward-difference Jacobian (step 1% of each parameter's range), halving a step up to five times if it does not lower the error. A fit ending on a bound is reported as such. Albedo, emissivity and per-pixel or per-block thermal inertia are not fitted: albedo trades off against g_{diff} at midday, and per-pixel values cannot transfer to unvisited pixels.

3 Numerical verification

Two sets of tests had to pass, in code, before the model was compared with any Mars data.

Analytic half-space. With the surface held at $T(0,t) = T_0 + \Theta\sin\omega t$, $\omega = 2\pi/P$, the periodic solution of (9) on a half-space is

$$T(z,t) = T_0 + \Theta\, e^{-z/\delta}\sin(\omega t - z/\delta), \tag{15}$$

so amplitude decays as $e^{-z/\delta}$ and phase lags by z/δ radians (Fig. 1).

Table 3 lists all checks. The forcing checks are a consistency test, not independent validation: the training table's flux and elevation columns were computed with the same constants and formulas. Diffuse light and sky infrared meet data only in the comparison with REMS (Section 4).

4 Calibration and validation against REMS

4.1 Data

The frozen training table has 15 523 REMS bins from 319 sols in 11 blocks of contiguous sols. Models are evaluated at bin centres. Three row groups were set aside before modelling: 29 sols

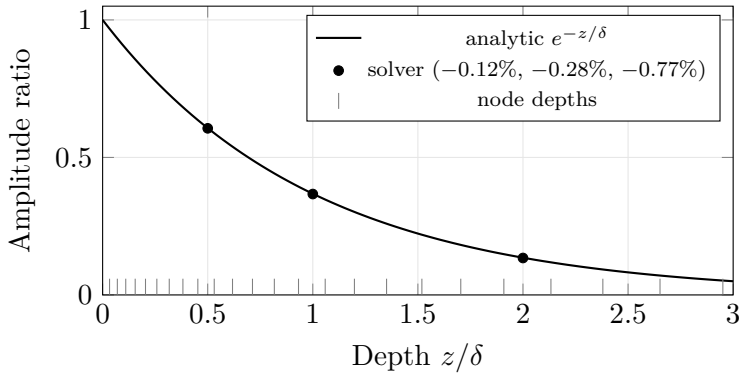


Figure 1: Half-space test. Diurnal temperature amplitude against depth: the analytic solution (line), the solver at 0.5δ, 1δ and 2δ (points; relative errors in the legend, too small to see at this scale), and the node positions of the $\gamma = 1.10$ grid down to 3δ (ticks).

Table 3: Verification results.

Check	Limit	Measured
Solver		
Half-space amplitude at 0.5δ / 1δ / 2δ	1%	−0.12% / −0.28% / −0.77%
Half-space phase lag	1%	within 0.04%
Change in surface T on halving Δt / every layer	0.1 K	0.032 K / 0.021 K
Sol-mean net surface flux, periodic state	$0.1\,\mathrm{W\,m^{-2}}$	$\leq 0.0002\,\mathrm{W\,m^{-2}}$
Spin-up to 0.01 K	50 sols	11–13 sols
Newton iterations per step (low and high τ)	20	≤ 2
Forcing		
Solar elevation vs. training table	0.5°	0.0008°
Direct flat and tilted flux vs. table ($> 20\,\mathrm{W\,m^{-2}}$)	1%	0.0063%

with a placeholder $\tau = 0.5$ (run and scored, never fitted); sols 3763–3778 (block 10), whose DTM slope at the rover, 23.9°, is suspect (scored with and without); and the sealed test, blocks 4 (sols 1343–1360) and 8 (sols 3069–3110), which the data loader drops unless explicitly told otherwise.

4.2 Protocol

Validation is leave-one-block-out over the nine training blocks (0, 1, 2, 3, 5, 6, 7, 9, 10). For held-out block b with n_b rows,

$$\mathrm{RMSE}_b = \left[\tfrac{1}{n_b}\textstyle\sum_{j\in b}\left(\hat{T}_j - T_j^{\mathrm{REMS}}\right)^2\right]^{1/2}, \qquad \mathrm{RMSE}_{\mathrm{macro}} = \tfrac{1}{9}\textstyle\sum_b \mathrm{RMSE}_b. \tag{16}$$

The macro average is the headline because block 0 (sols 1–170) holds about a quarter of the rows. All parameters, including the residual model, are refitted inside every fold. The **benchmark** is the mean REMS value per (15-minute LTST bin, 30° L_s bin) from each fold's training rows, falling back to the LTST-bin mean when a cell is empty. It is a seasonal lookup at the rover with no spatial information, so it cannot be a map, but it is a strong yardstick for skill in time. Any addition to the model had to improve the macro RMSE by 0.2 K and win in 5 of 9 blocks, a rule written before each experiment. Eleven configurations were tried in total (Section A). The sealed blocks were scored once, after every choice was frozen.

4.3 Physics only

At default parameters ($g_\mathrm{diff} = 0.5$, $f_\mathrm{IR} = 0.04$, $s_\mathrm{TI} = 1$) the model is about 18 K too cold at midday in every block, season and dust bin (macro RMSE 12.67 K). The deficit does not grow

Table 4: Input data. All layers are resampled onto the THEMIS 100 m template (1436 × 1477 pixels, 136.75°–139.24° E, 6.57°–4.15° S).

Quantity	Source	Notes
Ground, air temperature	REMS RDR [6, 7]	15-min LTST bins, mean of 1 Hz readings; training truth.
Thermal inertia	THEMIS, 100 m [4, 5]	Scaled by s_{TI} in the solver.
Albedo	TES MY24–26, ∼3 km [1, 24]	Three-year mean, bilinear; smooth trend only.
Elevation, slope, aspect	CTX stereo DTMs, 20 m [8]	Area-averaged to 100 m.
Dust optical depth	Mastcam, 880 nm [16, 15]	One value per sol.
Base imagery	CTX mosaic [17, 18]	Display only.

with dust (-16.7 K at $\tau \leq 0.4$; -14.1 K at $\tau = 1.3$–2), so excess extinction is not the cause. Pre-dawn bias grows with THEMIS thermal inertia (+13.6 K at 550–700 tiu, nearly all in block 0), which points to I being too high.

4.4 Fitted physics

Across folds, $g_{\mathrm{diff}} = 0.77$–0.88, $f_{\mathrm{IR}} = 0.027$–0.034 and $s_{\mathrm{TI}} = 0.70$–0.87 (all blocks: 0.82, 0.031, 0.74); s_{TI} sits on its lower bound in one fold. Fitting halves the error but does not reach the benchmark (Table 5). One g_{diff} cannot fit both seasons: the fitted model is 4–7 K cold at $L_{\mathrm{s}} = 30$–$120°$ and 2–5 K warm at 150–270°. Lowering the bound on s_{TI} to 0.5 changed no score by more than 0.04 K.

Table 5: LOBO RMSE (K) by held-out block: fitted physics (with its bias), benchmark, and physics plus residual (`flann_safe`; the map's configuration differs only by dropping its thermal-inertia terms, Section 4.6).

Block (sols)	Physics	Bias	Benchmark	Physics + residual
0 (1–170)	9.25	+4.06	5.04	7.46
1 (365–375)	4.05	−1.92	5.19	2.86
2 (584–865)	6.69	+0.42	5.59	5.36
3 (1160–1165)	6.07	−5.51	4.01	2.64
5 (1650–1865)	7.62	−5.69	5.54	5.54
6 (2550–2673)	7.90	−6.38	7.83	5.37
7 (2871–2968)	4.71	−0.96	4.33	4.33
9 (3550–3655)	6.94	+1.18	5.42	5.38
10 (3763–3778)	7.27	−6.51	5.51	3.72
Macro	6.72		5.38	4.74
Pooled, main rows	7.47		5.51	5.82

4.5 Thermal inertia seen by REMS

Scanning s_{TI} separately per block (a diagnostic; nothing enters the map) shows that block 0's night, day and diurnal-amplitude errors all point to the same scale (0.50, 0.60, 0.54). Sky infrared cannot substitute: block 0's best f_{IR} (0.030) equals the global fit, and repairing its nights with f_{IR} alone leaves a -8.8 K day bias. Multiplying each block's best scale by its THEMIS mean gives the thermal inertia REMS actually sees (Table 6).

Whatever THEMIS reports (386–637 tiu), REMS behaves like ground of 330–400 tiu. This agrees in magnitude with independent REMS analyses: 265–375 tiu for sols 1–100 [9]; 295 ± 37, 306 ± 38 and 452 ± 50 tiu at Rocknest, Point Lake and Yellowknife Bay [19]; 350–550 tiu apparent for bedrock-dominated surfaces [26]. Those studies attribute the orbital–in-situ difference to

Table 6: Best thermal-inertia scale per block (step 0.05) and the implied I.

Block	THEMIS mean (tiu)	All hours	Night	Day	Amplitude	Implied I (tiu)
0	637	0.55	0.50	0.60	0.54	350
1	463	0.80	0.85	0.80	0.78	370
2	458	0.80	0.75	0.85	0.77	367
3	465	0.75	0.95	0.60	0.76	349
5	386	0.85	1.00	0.75	0.77	328
6	501	0.80	1.00	0.60	0.73	401
7	473	0.80	0.80	0.75	0.75	378
9	399	0.90	0.85	1.00	0.93	360

footprint size and sub-pixel mixing of sand, loose material and bedrock [9, 26, 3], which varies from pixel to pixel. The literature thus supports $s_{\mathrm{TI}} < 1$ at the rover but not one scale for the whole crater. At its best scale, block 0 still peaks too hot at 11–14 h and cools too fast at 18–19 h, the signature of a thin low-inertia layer over denser ground [9, 23] that a homogeneous column cannot reproduce.

4.6 Residual model

The residual $y = T^{\mathrm{REMS}} - T_{\mathrm{base}}$ was first fitted with gradient-boosted trees, which beat the benchmark by only 0.10 K and could not extrapolate to block 0's thermal inertia (566–682 tiu, outside every other block's ≤ 514). We replaced them with a functional-link expansion [21] and a ridge penalty. With $\omega_d = 2\pi\mathrm{LTST}/24$ and $\omega_s = 2\pi L_{\mathrm{s}}/360$,

$$\boldsymbol{\varphi} = \{\tau\} \cup \{g(k\omega_d),\, \tau g(k\omega_d)\}_{k\leq 3} \cup \{g(m\omega_s),\, \tau g(m\omega_s)\}_{m\leq 2} \cup \{c,\, c\sin\omega_d,\, c\cos\omega_d\}_{c\in\mathcal{C}}, \quad (17)$$

with $g \in \{\sin, \cos\}$ and $\mathcal{C}$ the static terms, always including T_{base}. Columns are standardised, and $\hat{\boldsymbol{\beta}} = \arg\min\|\mathbf{y} - \Phi\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2$ with $\lambda = 1$, fixed. Because a linear model extrapolates, each covariate in $\{I_{\mathrm{THEMIS}}, s, A, T_{\mathrm{base}}\}$ is clamped to the 1st–99th percentiles of the fold's training rows, both when fitting and when predicting, so LOBO scores exactly the computation run on the map.

Albedo was dropped: the rover's albedo spans only 0.24–0.26, yet its coefficient moved the residual by 29 K over the crater's 0.10–0.35, acting as a campaign identifier. The resulting model (`flann_safe`) passed the rule with macro 4.74 K, better than the trees in 8 of 9 blocks. Its thermal-inertia terms, however, earned only 0.12 K at the rover while shifting 92% of map pixels by more than 2 K at night, because 82% of crater pixels lie below the rover's clamp minimum of 361 tiu. The map therefore uses the same model without them (`flann_safe_no_ti`: macro 4.86 K, pooled 5.83 K). That choice was made after the first result was seen and is reported as such. A warm night bias of +1.8 to +2.5 K remains at 22–03 h.

4.7 Near-surface air temperature

REMS air is about 8 K warmer than the ground at night and 26 K colder at 11–12 h. The air model uses basis (17) with $\mathcal{C} = \{\hat{T}_{\mathrm{ground}}\}$, the out-of-fold predicted ground temperature, so its score reflects what the map has; elevation is excluded because 83% of crater pixels lie above the rover's highest position. It scores 4.05 K macro against the benchmark's 4.04 K (better in 4 of 9 blocks), fails the adoption rule, and is published as the better of two configurations with the shortfall stated. Block 0's predicted ground is 6.9 K too warm at night, and every model trained on it learns to ignore night ground (0.02–0.08 K of air per K of ground at 00–05 h, against 0.13–0.23 K/K when fitted to observed ground). Night air on the map is therefore nearly uniform (spatial standard deviation about 0.1 K at 02:00) because of a ground-model error, not physics.

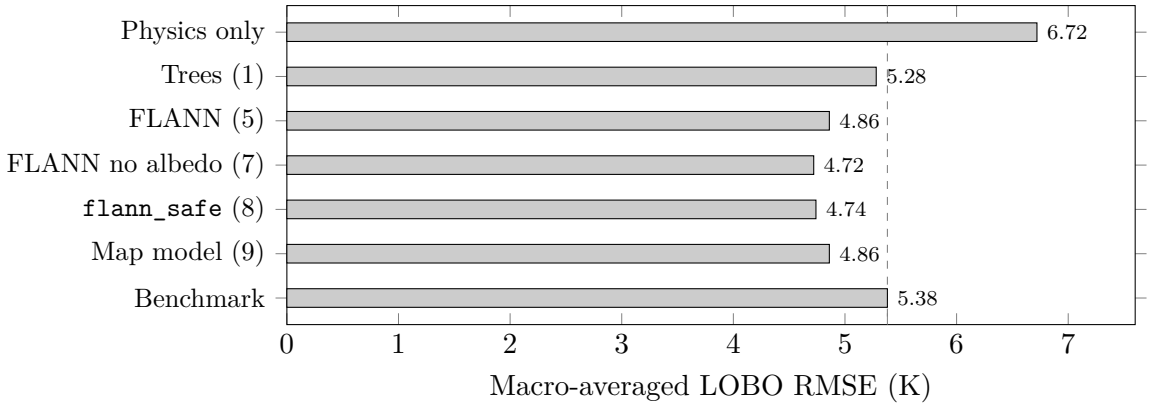


Figure 2: Ground-temperature macro RMSE under LOBO validation; dashed line, benchmark. Numbers in parentheses are configuration numbers (Section A).

4.8 Sealed test

Table 7: Sealed test, scored once. RMSE (bias) in K.

	Block 4	Block 8	Sealed macro	LOBO macro
Ground model (map)	4.33 (+2.09)	6.18 (−0.84)	**5.25**	4.86
Ground benchmark	4.79 (+1.28)	5.32 (+0.07)	**5.06**	5.38
Air model (map)	3.30 (+0.67)	4.90 (−3.18)	**4.10**	4.05
Air benchmark	3.92 (+1.56)	4.46 (−1.16)	**4.19**	4.04

The ground model's LOBO lead did not hold (Table 7): it won block 4 and lost block 8, where it was 2.7–3.3 K too cold at 13–15 h. With two blocks and 25 sols, neither difference is beyond noise. Both blocks lie within 1 km of training positions, so the test says nothing about other terrain.

5 Application 1: temperature maps

The static layers are block-averaged to a 718×738 grid at about 200 m (529 884 pixels, 76 nodata). The model runs at $L_s = 15$, 75, 195, 255 and 340°, with τ the median Mastcam value within ±15° of each (0.78, 0.49, 0.69, 1.09, 1.11), and writes 48 frames per season at 30-minute intervals. The five seasons ran as parallel processes in chunks of 1000 columns, in 7769 s of wall time (13.9–14.7 ms per column per worker). Larger chunks were slower (38 ms per column at 10 000) because the work arrays no longer fit in cache. Spin-up took 11–14 sols per chunk everywhere, the same as at the rover.

Each map is written as a cloud-optimised GeoTIFF, a 16-bit PNG, a colour PNG (150–320 K) and raw `uint16`, with a manifest recording the code commit. The map code reproduces the rover-pixel model to within 0.0066 K; decoded files agree with the float32 master to 0.005 K (quantisation) or half a colour step; and georeferencing was confirmed in QGIS.

Slope enters the residual clamped to 1.25–12.27° (28.4% of pixels clamped); the physics sees the true slope, so steep crater walls and Mount Sharp's flanks rest on the baseline alone. At $L_s = 75°$, 24% of pixel-frames have predicted ground below the rover's training range.

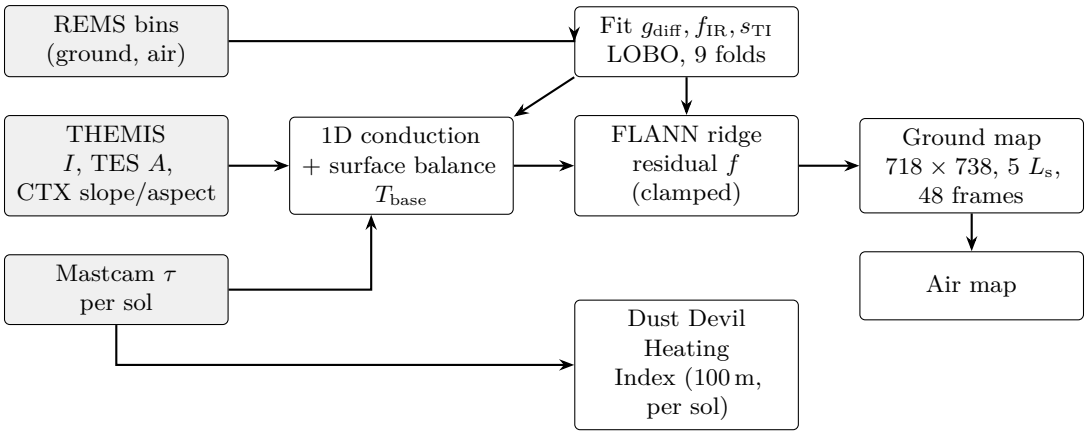


Figure 3: Data flow. The heating index uses the same static inputs, dust and solar geometry as the thermal model but no fitted parameter.

Table 8: Ground-temperature map statistics (K). "Clamped": fraction of pixel-frames where $T_{\rm base}$ was clamped as a residual input.

$L_{\rm s}$	τ	Min	p1	p99	Max	Residual	Clamped
15	0.78	169.7	181.3	282.6	292.9	−11.4 to +18.1	10.3%
75	0.49	151.5	167.8	270.5	279.7	−10.4 to +21.5	31.2%
195	0.69	175.4	185.9	302.2	313.1	−17.2 to +14.9	18.4%
255	1.09	181.6	189.9	298.6	310.1	−18.8 to +16.0	18.8%
340	1.11	178.6	188.5	288.8	299.8	−15.1 to +18.3	9.5%

6 Application 2: Dust Devil Heating Index

6.1 Physical basis

Dust devils form when the ground becomes much hotter than the air above it, so that a shallow unstable layer rises and begins to rotate. The ground–air gap grows with the sunlight a surface absorbs and shrinks with its thermal inertia: a low-inertia surface cannot conduct heat away quickly, so its skin heats more.

The model of Section 2 makes this quantitative. Linearise (9) on a half-space driven by a periodic surface heat flux $F = \mathrm{Re}\{\hat{F} e^{i\omega t}\}$. The solution $T = \mathrm{Re}\{\hat{\Theta} e^{-(1+i)z/\delta} e^{i\omega t}\}$ gives a surface flux $k(1+i)\hat{\Theta}/\delta$, so

$$|\hat{\Theta}| = \frac{|\hat{F}|\,\delta}{\sqrt{2}\,k} = \frac{|\hat{F}|}{I\sqrt{\omega}}, \tag{18}$$

using $\delta = \sqrt{2\kappa/\omega}$ and $k/\sqrt{\kappa} = I$. In this linear limit the surface temperature swing scales as $1/I$; radiative loss, which the linearisation omits, weakens that dependence. We adopt a $1/\sqrt{I}$ weighting as a heuristic between the two limits, keeping the contrast between sand and bedrock without letting low-inertia outliers dominate. It is an assumption, not a derived result.

6.2 Definition

For each 100 m pixel and each quarter-hour LTST bin t,

$$p(\mathbf{x}, t) = \frac{(1 - A(\mathbf{x}))\, Q(\mathbf{x}, t)}{\sqrt{\max\{I_{\rm THEMIS}(\mathbf{x}),\, 50\}}}, \tag{19}$$

where Q is the tilted direct beam (5) plus a fixed diffuse term $0.15\, S_0 \mu_0 (1 - e^{-\tau}) \frac{1}{2}(1 + \cos s)$, with S_0 scaled from 589 W m^{-2} at the semi-major axis. These coefficients are fixed, not fitted,

and differ slightly from the thermal model's. The score, its daily mean over 9–16 h (28 bins) and its daily maximum are

$$\sigma(\mathbf{x},t) = \min\Big\{\frac{p}{p_{\mathrm{ref}}},\,1\Big\}\,e^{-(s/10^\circ)^2}, \qquad \bar{\sigma}(\mathbf{x}) = \tfrac{1}{28}\textstyle\sum_{t\in[9,16]}\sigma, \qquad \sigma_{\max}(\mathbf{x}) = \max_t \sigma. \tag{20}$$

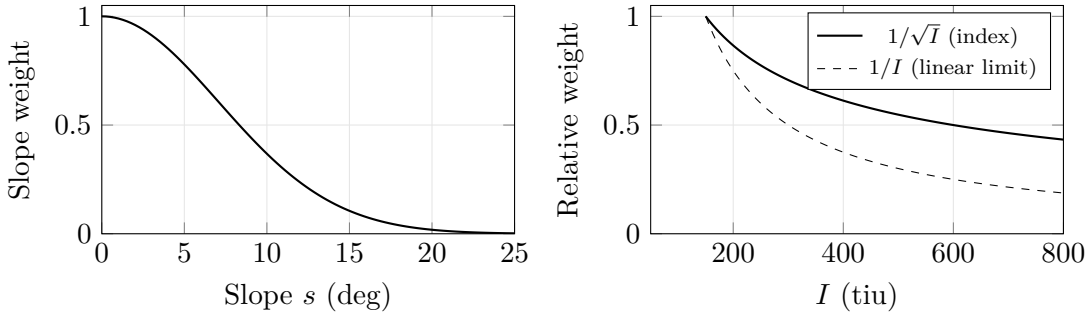


Figure 4: The two terrain weights in the heating index: the slope penalty $e^{-(s/10^\circ)^2}$ (left), and the thermal-inertia weighting normalised at 150 tiu, compared with the linear half-space limit (18) (right).

Maps use a fixed grid of 96 bin centres, independent of each sol's REMS coverage. An earlier design that used each sol's own REMS bins let coverage gaps (16–21 bins between 9 and 16 h) drive day-to-day changes more than dust did (largest day-to-day change in the daily mean 0.067, against 0.012 on a common grid).

6.3 A shared reference across sols

The normaliser p_{ref} is the 99.5th percentile of p over all pixels and all 96 bins of one reference sol, and is used for every sol so that a dusty or low-sun sol scores lower than a clear one. The statistic varies about twofold across the 263 sols with at least 40 REMS bins (10.4 to 23.1). The first choice, sol 800 ($\tau = 1.5$), was the 12th dimmest, and 243 of 263 sols then had more than 20% of pixels at the cap. Sol 1850 ($L_{\mathrm{s}} = 76^\circ$, measured $\tau = 0.44$) sits at the median and was adopted ($p_{\mathrm{ref}} = 16.47$); candidates were limited to sols with measured τ within one sol, 55–96 bins and rover correlation ≥ 0.93, and the sealed blocks were excluded. Three fixed levels are cut at the top 50, 30 and 10% of the reference sol's $\bar{\sigma}$: low $[0.348, 0.474)$, moderate $[0.474, 0.590)$, high ≥ 0.590.

Dust optical depth is interpolated from Mastcam measurements within five sols. Where none exists (sols 1–32 and some gaps), τ is the median of Mastcam sols within $\pm 10^\circ$ of L_{s} in other Mars years, excluding the 2018 global storm (sols 2082–2133) and requiring at least five sols. Every output records its τ source.

6.4 Results and in-situ check

At the rover pixel, the per-bin proxy is compared with the REMS ground-minus-air gap through the Pearson correlation and a max/max scale (about 2.7 K per proxy unit on sol 1850, so a score of 1.0 corresponds to a gap of roughly 45 K). The scale enters only this check, never the maps.

For a crater-floor pixel (slope 1°, $A = 0.14$, $I = 363$ tiu), mean sunlight over 9–16 h falls from 219 W m^{-2} on sol 1850 to 174 W m^{-2} on sol 800, and the index from 0.591 (high) to 0.469 (moderate). Across runs, dusty sols put only 0.2–0.4% of the crater in the high level, against 6.4–10% on clear sols (Table 9).

The correlation shows that the index follows the diurnal heating cycle at one pixel. Any insolation-driven quantity shares that cycle, so a high r is necessary but is not evidence of spatial skill or of dust devil occurrence. Because τ is one value per sol for the whole crater, dust scales

Table 9: Heating index by run: share of crater pixels per level, and rover correlation.

Run (sols)	τ	Low	Moderate	High	Rover r
Clear, 1830–1865	0.44–0.51	20.0–22.2%	19.3–20.0%	6.4–10.0%	0.84–0.97
Dusty, 799–804	1.44–1.52	27.2–27.4%	6.7–9.7%	0.2–0.4%	0.94–0.98
Climatology τ (7 sols)	0.48–0.80	16.4–22.5%	19.2–19.8%	5.9–17.3%	0.80–0.97

the map but cannot move where activity is favoured (sol 800 and sol 1850 maps correlate at 0.996). About 100 of 263 sols would still have more than 20% of pixels at the cap against sol 1850. The index has not been compared with observed dust devils; Curiosity's Navcam vortex surveys and the pressure drops in 1 Hz REMS data [12] are the natural tests.

7 Discussion

7.1 Comparison with KRC

Table 10: Design comparison with KRC as described in [14].

Aspect	KRC	This model
Atmosphere	One layer, delta-Eddington, energy-conserving	Closed-form extinction, diffuse and sky-IR terms; two fitted coefficients
Layering	Geometric thickening with depth	Geometric, $\delta/30$ to 11δ, $\gamma = 1.10$
Time integration	Explicit; time step doubled with depth	Implicit (backward Euler), one Δt; Newton on surface
Materials	Homogeneous or two zones; optional temperature dependence	Homogeneous, constant properties
Seasonal effects	Multi-year runs; CO_2 frost	No frost; periodic single-sol runs per L_s
Calibration	Forward model	Fitted to REMS, plus statistical residual

Table 10 summarises the differences. The model here is deliberately simpler in its physics and deliberately more empirical in its calibration. The simplification suits Gale's equatorial, frost-free surface and the diurnal question asked, but it costs physical meaning: g_{diff} and f_{IR} absorb whatever the parameterised atmosphere misses, including sensible heat exchange, so they should not be read as radiative-transfer quantities. Periodic single-sol runs also ignore the seasonal heat wave below the diurnal skin depth, which multi-year runs capture.

7.2 Limitations

1. **Spatial skill is unmeasured.** Every REMS position lies within about 10×5 km, and the sealed blocks within 1 km of training positions.
2. **The thermal-inertia scale is not supported away from the rover.** The rover samples the high end of the crater's thermal inertia (crater median 281 tiu, rover median 470 tiu). With $s_{\mathrm{TI}} = 0.74$, a 281 tiu pixel runs at 208 tiu, and no data test I below about 330 tiu.
3. **Homogeneous soil.** Block 0's residual diurnal shape error indicates layering the model cannot represent.
4. **Night-time air has almost no spatial structure,** for the reason given in Section 4.
5. **Model ranking.** Eleven configurations were tried on nine blocks, and the sealed result reversed the ground model's LOBO lead; differences of a few tenths of a kelvin are within noise.

6. **The heating index is one ingredient.** Loose dust, ambient wind and boundary-layer depth are not modelled, and its weights are assumptions.

7.3 Further work

In order of expected value: a thermal-inertia mapping that shrinks THEMIS variation toward the in-situ value instead of scaling it; a two-layer regolith; a one-off comparison with the Mars Climate Database at the Gale cell, or with KRC run at the rover pixel, as an independent check; and validation of the heating index against Navcam surveys and REMS pressure-drop detections.

8 Conclusions

A vectorised implicit thermal model with three fitted parameters and a clamped ridge residual maps ground and air temperature across Gale Crater at 200 m. The solver meets its analytic, convergence and energy tests by wide margins; the remaining error lies in forcing, parameters and missing physics. At the rover the ground model improves on a seasonal lookup under block-held-out validation but not on a sealed test, and the air model is level with it. The clearest scientific result is that REMS sees ground of 330–400 tiu along the whole traverse whatever THEMIS reports, consistent with independent REMS analyses whose explanation, footprint mixing, does not support a single crater-wide correction. The Dust Devil Heating Index gives a measure of solar-driven surface heating that is comparable across sols and tracks the in-situ ground–air gap; whether it predicts dust devils remains to be tested.

Data and code availability

Code: the `heatmap` folder of `MarsDataPipline` (solver, residual model, map export, ten driver notebooks, 13 gate tests) and the `dust_devil` folder (pipeline notebook, 70 tests) in the team repository. Map products, inputs and score tables are on the project's shared drive. Map manifests record code commit `5784569`.

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A Configuration register

#	Configuration	Outcome
1	Trees, static covariates without elevation	Adopted at first; superseded
2	Trees, all covariates	Elevation rejected (+0.07 K, bar 0.2 K)
3	Trees, time only	0.34 K worse than 1
4	Trees, 1 + absorbed flux	0.04 K worse than 1
5	FLANN ridge	Passed rule; unsafe to extrapolate
6	FLANN, clamped	Clamping costs 0.02 K
7	FLANN, no albedo	Albedo acted as a campaign identifier
8	FLANN, no albedo, clamped (`flann_safe`)	Passed rule vs 1 (+0.54 K, 8/9 blocks)
9	8 without thermal-inertia terms	Used on the map
10	Air, hybrid	Failed rule; published as better of two
11	Air, time only	Lost to benchmark

B Reproducibility

- Environment: virtual environment `gale-env` (NumPy 2.5.4, pandas 3.1.0, scikit-learn 1.9.1, rasterio 1.5.2, Pillow 13.0.0); the solver depends on NumPy only.
- Tests: `pytest heatmap` (13 tests) and `pytest dust_devil/tests` (70 tests).
- Notebook order (`heatmap/notebooks/`): `baseline_diagnostic`; `baseline_fit_lobo`; `residual_lobo`; `residual_flann_lobo`; `ti_sensitivity_lobo`; `block0_diagnostic`; `map_run`; `air_lobo`; `air_map_run`; `sealed_test`. Dust devil: `dust_devil/dust_devil_pipeline.ipynb`.
- Fixed before results were seen: folds, exclusions, benchmark, decision rules, ridge penalty, FLANN basis, clamp percentiles, sealed blocks.
- Chosen after results were seen, and disclosed: dropping albedo terms (configurations 7–8) and thermal-inertia terms (9); the 150–320 K colour scale; the heating-index reference sol and level cuts.